1 Mathematical Formulation

1.1 Original System Model

Sets

$$
\begin{aligned}
&\mathcal{D} = \{\mathrm{ER}, \mathrm{SURG}, \mathrm{ICU}, \mathrm{SD}\} \\
&\mathcal{T} = \{1, \ldots, T\} \\
&\mathcal{A} = \{(i,j) : i,j \in \mathcal{D},\ \text{feasible patient transitions}\}
\end{aligned}
$$

Parameters

$$
\begin{aligned}
R_d &: \text{ room capacity in department } d, \\
S_d &: \text{ permanent staff capacity in department } d, \\
\gamma_d &: \text{ ready-out discharge limit per hour in } d, \\
c_d^{wait} &: \text{ waiting cost per patient-hour in } d, \\
c^{temp} &: \text{ temporary staff cost per unit}, \\
c^{div} &: \text{ ambulance diversion cost}, \\
W_{dt} &: \text{ walk-in arrivals to } d \text{ at time } t, \\
A_t &: \text{ ambulance arrivals at time } t.
\end{aligned}
$$

Decision Variables

$$
\begin{aligned}
x_{dt} &: \text{ patients in department } d \text{ at end of time } t, \\
w_{dt} &: \text{ patients waiting in } d \text{ at end of time } t, \\
u_{dt} &: \text{ patients admitted into } d \text{ at time } t, \\
f_{ijdt} &: \text{ patients transferred from } i \text{ to } j \text{ at time } t, \\
y_{dt} &: \text{ temporary staff in } d \text{ at time } t, \\
a_t &: \text{ ambulance patients admitted at time } t, \\
v_t &: \text{ ambulance patients diverted at time } t.
\end{aligned}
$$

Objective

$$
\min \sum_{t \in \mathcal{T}} \left[\sum_{d \in \mathcal{D}} \left(c_d^{wait} w_{dt} + c^{temp} y_{dt} \right) + c^{div} v_t \right]. \tag{1}
$$

Constraints Ambulance balance:

$$
a_t + v_t = A_t, \quad \forall t. \tag{2}
$$

Admission feasibility:

$$
u_{ER,t} \le w_{ER,t-1} + W_{ER,t} + a_t, \tag{3}
$$

$$
u_{dt} \le w_{d,t-1} + W_{d,t}, \quad \forall d \ne \mathrm{ER}. \tag{4}
$$

Flow conservation (patients):

$$x_{dt} = x_{d,t-1} + u_{dt} + \sum_{i:(i,d)\in\mathcal{A}} f_{ijdt} - \sum_{j:(d,j)\in\mathcal{A}} f_{djdt}, \quad \forall d,t. \tag{5}$$

Ready-out limits:

$$\sum_{j:(d,j)\in\mathcal{A}} f_{djdt} \le \gamma_d, \quad \forall d,t. \tag{6}$$

Room capacity:

$$x_{dt} \le R_d, \quad \forall d,t. \tag{7}$$

Staffing feasibility (1:1 ratio):

$$x_{dt} \le S_d + y_{dt}, \quad \forall d,t. \tag{8}$$

Waiting update:

$$w_{ER,t} = w_{ER,t-1} + W_{ER,t} + a_t - u_{ER,t}, \tag{9}$$

$$w_{dt} = w_{d,t-1} + W_{d,t} - u_{dt}, \quad \forall d \ne \text{ER}. \tag{10}$$

1.2 Optimized System Model (with Triage)

The optimized model extends the baseline formulation by introducing a triage buffer layer, patient classification, admission caps, and controlled ER double-bunking.

Additional Parameters

$$\begin{aligned} R^{tri} &: \text{ triage bed capacity,} \\ S^{tri} &: \text{ permanent triage staff,} \\ c^{triwait} &: \text{ triage waiting cost,} \\ c^{double} &: \text{ double-bunk congestion penalty,} \\ \bar{u}_d &: \text{ admission cap per hour in } d. \end{aligned}$$

Additional Decision Variables

$$\begin{aligned} z_t &: \text{ patients in triage beds,} \\ \tilde{w}_t &: \text{ patients waiting in triage,} \\ \ell_t &: \text{ low-acuity patients released,} \\ h_t^{ER} &: \text{ high-acuity routed to ER,} \\ h_t^{dir} &: \text{ high-acuity routed directly to ICU/SURG,} \\ y_t^{tri} &: \text{ temporary triage staff,} \\ \delta_t &: \text{ ER overflow above base capacity.} \end{aligned}$$

Extended Objective

$$\min \sum_{t \in \mathcal{T}} \left[\sum_{d \in \mathcal{D}} \left(c_d^{wait} w_{dt} + c^{temp} y_{dt} \right) + c^{triwait} \tilde{w}_t + c^{temp} y_t^{tri} + c^{double} \delta_t \right]. \tag{11}$$

Additional Constraints Triage classification:

$$\ell_t + h_t^{ER} + h_t^{dir} = z_t, \quad \forall t. \tag{12}$$

Admission caps:

$$u_{dt} \le \bar{u}_d, \quad \forall d, t. \tag{13}$$

Controlled ER double-bunking:

$$x_{ER,t} \le 2R_{ER}, \quad \forall t. \tag{14}$$

Overflow tracking:

$$\delta_t \ge x_{ER,t} - R_{ER}, \quad \forall t. \tag{15}$$

Triage staffing:

$$z_t + \tilde{w}_t \le 3(S^{tri} + y_t^{tri}), \quad \forall t. \tag{16}$$

Generated from ‘1.ipynb‘ model code: original and optimized triage-based implementations.